# SEMICONDUCTORS

# Semiconductor

### Definition:

A substance whose electrical conductivity lies between conductor and insulator is called semiconductor.

# Types of Semiconductors

### Intrinsic Semiconductor:

A pure semiconductor without impurities is called intrinsic semiconductor.

### Extrinsic Semiconductor:

A semiconductor doped with impurities is called extrinsic semiconductor.

# Doping

### Definition:

The process of adding impurities to a pure semiconductor is called doping.

## Energy Bands

- Valence Band
- Conduction Band
- Forbidden Energy Gap

## P-type Semiconductor

### Definition:

A semiconductor formed by adding trivalent impurity is called P-type semiconductor.

### Majority Charge Carriers:

Holes are the majority charge carriers.

## N-type Semiconductor

### Definition:

A semiconductor formed by adding pentavalent impurity is called N-type semiconductor.

### Majority Charge Carriers:

Electrons are the majority charge carriers.

# PN Junction

## Definition:

The junction formed by joining P-type and N-type semiconductors is called PN junction.

## Depletion Region:

The region near PN junction depleted of free charge carriers is called depletion region.

# Semiconductor Diode

## Definition:

A PN junction device used for rectification is called semiconductor diode.

## Forward Bias

In forward bias, the P-side is connected to the positive terminal and N-side to the negative terminal.

**Result:**

Current flows easily through diodes.

## Reverse Bias

In reverse bias, the P-side is connected to the negative terminal and N-side to the positive terminal.

**Result:**

Very small current flows through the diode.

## Rectifier

A device which converts alternating current into direct current is called a rectifier.

## LED

Light Emitting Diode emits light when current passes through it.

# Transistor

### Definition:

A semiconductor device used for amplification and switching is called a transistor.

## Types of Transistors

- NPN Transistor
- PNP Transistor

## Logic Gates

Logic gates are electronic circuits used to perform logical operations.

### Basic Gates:

- AND Gate
- OR Gate
- NOT Gate

## Quick Revision Points

- Semiconductor conductivity lies between conductor and insulator.
- Doping increases conductivity.
- P-type has holes as majority carriers.
- N-type has electrons as majority carriers.
- PN junction forms a depletion region.
- Diode is used for rectification.
- LED emits light.
- The transistor is used for amplification.

## Important Terms

- Intrinsic Semiconductor
- Extrinsic Semiconductor
- PN Junction
- Forward Bias
- Reverse Bias
- Rectifier
- LED
- Transistor